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Interactive Risk (IR): An omnidirectional safety metric of CAVs based on multimodal trajectory prediction and driving risk field

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ABSTRACT

Traffic accidents pose a significant threat to human life and property, and with the increasing presence of connected and autonomous vehicles (CAVs), effective risk assessment has become more critical. Current safety metrics, often limited to longitudinal or lateral assessments, fail to address omnidirectional risks or account for the uncertainties associated with vehicle intentions. This paper introduces a new omnidirectional safety metric, Interactive Risk (IR), which combines the concept of the driving risk field with multimodal trajectory prediction. IR captures the uncertainty of vehicle intentions, quantifies the probability and severity of potential accidents, and provides a comprehensive measure of traffic risk. Through case studies of typical collision scenarios and experiments with the simulation and real world dataset, we demonstrate that IR accurately reflects the risk levels faced by CAVs, detects collision risks earlier, and aligns more closely with human intuition compared to baseline safety metrics. Furthermore, we propose four key applications of IR, including traffic risk monitoring, ego-vehicle risk warning, driving decision-making performance evaluation, and motion and trajectory planning. The results highlight the potential of IR to enhance safety assessment in dynamic traffic environments and provide valuable insights for future research and application in autonomous vehicle systems.

1. Introduction

Traffic accidents pose a significant threat to human life and property, resulting in 1.19 million fatalities globally in 2023 (Anon, 2023). With the increasing presence of connected and autonomous vehicles (CAVs) as well as human-driven vehicles (HDVs), the need for effective risk assessment in dynamic traffic environment has become more critical than ever (Wang et al., 2020). Safety metrics play a vital role in facilitating this assessment (Tafidis and Pirdavani, 2023); if these metrics can accurately evaluate the risks under various hazardous scenarios, they have the potential to significantly reduce or mitigate the occurrence of traffic accidents.

Current traffic safety metrics are typically categorized into longitudinal and lateral assessments. Common longitudinal metrics include Time to Collision (TTC) (Hayward, 1972), Time Headway (THW) (Vogel, 2003), Time to Reaction (TTR) (Hillenbrand et al., 2006), Time Integrated Time-to-Collision (TIT) (Minderhoud and Bovy, 2001), and combinations of TTC and THW (Kondoh et al., 2008). For lateral risk assessment, metrics such as Time to Lane Cross (TLC) (Mammar et al., 2004), Car's Current Position (CCP) (Risack et al., 2000), and Future Offset Distance (FOD) (Xu et al., 2014) are frequently used. These metrics demonstrate effectiveness within their respective contexts. For instance, TTC is widely utilized in Advanced Driver Assistance Systems (ADAS) and has proven instrumental in reducing rear-end collisions. However, when longitudinal and lateral risks are coupled, these metrics often fail to deliver satisfactory results.

Developing safety metrics that integrate both longitudinal and lateral aspects is important. Current research categorizes vehicle conflicts or collisions into four types: rear-end, sideswipe, angle, and head-on Kaplan and Prato (2012). Among these, sideswipe and angle collisions are particularly challenging to accurately quantify using solely longitudinal or lateral metrics. In fact, there are some existing omnidirectional safety metrics, primarily utilizing two approaches: combining longitudinal and lateral metrics (Laureshyn et al., 2010; Guo et al., 2023) or employing a vectorized version of TTC (Hou et al., 2014; Ward et al., 2015;

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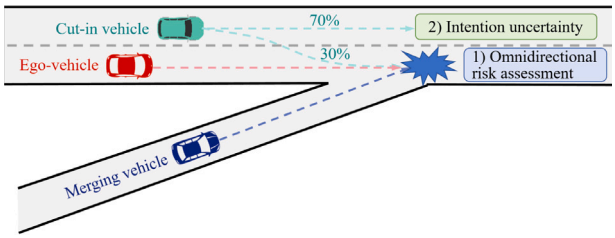


Fig. 1. Two main challenges faced by existing traffic safety metrics.

Venthuruthiyil and Chunchu, 2022). However, these omnidirectional metrics face two main issues: (1) the risk coupling mechanisms, that is, the strategies used to integrate longitudinal and lateral risk factors, are often simplistic, making it difficult to achieve effective results; and (2) they typically rely on kinematic or dynamic methods, which fail to account for the intentions and uncertainties of drivers and CAVs.

The use of field offers a promising approach to addressing omnidirectional risk assessment, as it can naturally represent the uncertainty of potential future risks. Some existing studies have employed the concept of field to develop corresponding safety metrics (Wang et al., 2015a, 2016; Li et al., 2020a). However, the current modeling of driving risk-related fields remains primarily based on vehicle kinematics, constructed from the current state of the vehicle (such as position, velocity, and acceleration). As a result, these models still do not account for the intentions of drivers or vehicles. To sum up, existing traffic safety metrics face two main challenges: (1) insufficient research on omnidirectional safety metrics and (2) a lack of consideration for uncertainties arising from intentions, as illustrated in Fig. 1.

Data-driven vehicle trajectory and intention prediction have developed rapidly in recent years (Zhou et al., 2023; Lan et al., 2023; Cheng et al., 2023). Combining multimodal trajectory prediction, which accounts for CAVs intentions, with the omnidirectionality of field in risk assessment holds potential to address the limitations of current safety metrics mentioned above. This is also the focus of this research. In our previous work (Jiang et al., 2024), we have explored this area by combining Gaussian distribution with multimodal trajectory prediction to establish the Driving Risk probability (DRP) model and propose a risk quantification model based on DRP. However, this work does not consider the direction of risk, resulting in an incomplete modeling of interactive risk. Therefore, a deeper exploration of the mechanisms underlying risk generation is needed to better align with human subjective perceptions of risk.

The main contributions of this paper include:

1. We propose a new driving risk assessment model — Integrated Driving Risk Field (IDRF) — by combining multimodal trajectory prediction with the concept of field. IDRF captures the uncertainty of vehicle intentions while quantifying the direction of risks generated by vehicles.
2. Based on IDRF, we introduce a new omnidirectional safety metric — Interactive Risk (IR), which quantifies the probability and severity of accidents to measure the level of risk.
3. We validate the effectiveness of the proposed safety metric through case studies of typical collision scenarios, as well as evaluations on both simulated and real-world driving data from publicly available datasets.

The remainder of this paper is organized as follows. In Section 2, a comprehensive literature review of the related research is conducted. Section 3 describes the modeling methodology of IR in details and Section 4 presents the verification of it through different aspects. Section 5 proposes four key applications of IR, while Section 6 concludes the paper and outlines future work.

2. Related works

In Section 1, we briefly introduce some commonly used safety metrics. In this section, we will provide a detailed overview of the work relevant to our research, including current longitudinal and lateral risk assessment methods, existing approaches for coupled longitudinal-lateral risk assessment, a general classification of risk assessment methods, risk evaluation methods based on field, and data-driven trajectory prediction techniques.

In longitudinal risk assessment, indicators can be categorized into time-based, distance-based, and acceleration-based metrics.

Time-based metrics. TTC (Hayward, 1972) is the most commonly used among the time-based metrics, which refers to the time remaining for a collision to occur if two vehicles continue to travel at their current velocities. In car-following scenarios where the relative velocity between two vehicles is low, the value of TTC tends to be high; therefore, the inverse of TTC, denoted as TTCi (Fancher et al., 2001), is also frequently used to reflect the collision risk between the vehicles. Another commonly used metric is THW (Vogel, 2003), which indicates the time difference between the vehicles passing the same point. When both vehicles are traveling at the same velocity, TTC may not effectively measure potential collision risk, making THW a useful supplementary metric. Additionally, TTR (Hillenbrand et al., 2006) is another time-based metric that represents the time available for a driver to execute emergency maneuvers, such as braking, accelerating, or steering, in response to a hazardous situation.

Distance-based metrics. Classical safety distance models include the Mazda model (Doi et al., 1994), Honda model (Fujita et al., 1995), and Berkeley model (Seiler et al., 1998), among others. These models share a similar structure, combining vehicle velocity, braking deceleration, and reaction time to estimate the minimum distance required for safe braking.

Acceleration-based metrics. This class of indicators focuses on the required deceleration to match the speed of a preceding vehicle and avoid a collision (Jansson, 2005). A higher required deceleration indicates a more critical situation and a higher collision risk.

These longitudinal safety metrics can be used individually or in combination. They are particularly effective in assessing and preventing rear-end and head-on collisions.

Lateral risk assessment can similarly be divided into time-based and distance-based metrics.

Among the time-based metrics, TLC (Mammar et al., 2004) is a classic measure, representing the time it takes for the vehicle's front wheels to cross the lane boundary from its current position.

For distance-based metrics, CCP (Risack et al., 2000) assesses the risk of lane departure by measuring the distance between the vehicle's front wheels and the lane boundary; the smaller the gap, the higher the risk. FOD (Xu et al., 2014) evaluates risk by determining whether the vehicle has exceeded a virtual lane boundary. A key advantage of FOD is that the virtual boundary can be set inside or outside the real lane line based on the driver's preferences, allowing for adaptation to driving habits. Distance-based metrics also include the acceptable gap model (Toledo et al., 2003). Here, the gap refers to (1) the distance between the front of the vehicle and the rear of the lead vehicle in the target lane, and (2) the distance between the rear of the vehicle and the front of the following vehicle in the target lane during lane-changing maneuvers. Critical gaps are calculated based on traffic flow conditions, and if the actual gap is larger than the critical gap, a lane change is considered acceptable.

It is evident that current lateral safety metrics primarily focus on lane-changing operations. If there are no lane markings, many safety metrics lose their effectiveness. Furthermore, these metrics are insufficient for quantifying other lateral risks, such as potential collisions involving vehicles from different directions at intersections or merging vehicles.

As mentioned earlier, current omnidirectional risk assessment methods can generally be divided into two categories: combination of longitudinal and lateral metrics, and vectorized version of TTC.

For the first category, some methods integrate TTC with additional measures, such as time advantage, to allow a smoother evaluation of collision risks during transitions between rear-end and angle conflicts (Laureshyn et al., 2010). Another approach applies a two-dimensional TTC, where separate TTC values are calculated for both the longitudinal and lateral directions, and the smaller value is selected to represent the most critical risk (Guo et al., 2023).

In the second category, one method is to calculate TTC by solving two-dimensional distance equations (Hou et al., 2014). Other strategies extend TTC by using vector-based calculations, determining collision time for vehicles that are not restricted to specific motion (Ward et al., 2015). Additionally, there is the concept of ACT, which generalizes TTC to a two-dimensional context and incorporates the potential evasive actions of vehicles (Venthuruthiyil and Chunchu, 2022).

It is clear that these metrics, rather than being truly omnidirectional safety metrics, are more accurately described as combinations of metrics applied in different directions.

In summary, the safety metrics mentioned above — whether lateral, longitudinal, or a combination of both — can be classified into two main categories: time-based metrics and kinematic-based metrics (such as distance and acceleration). In fact, there has been research focused on categorizing traffic safety metrics. For example, in Li et al. (2020b), risk assessment metrics are divided into five types. Beyond time-based and kinematic-based metrics, there are also statistics-based metrics (such as collision probability and machine learning methods) (Jansson, 2005; Wang et al., 2015b; Wu et al., 2023), potential field-based metrics (Wang et al., 2015a, 2016), and unexpected driving behavior-based metrics (like unusual driving events and traffic conflicts) (Worrall et al., 2010). Among these, the use of potential fields as a foundation for risk assessment has attracted increasing attention in recent years.

The artificial potential field (APF) method is a classical approach based on field theory, where the environment is modeled as a virtual field composed of attractive and repulsive forces to guide agent motion. It aligns well with the concept of omnidirectional risk assessment. Research on APF began in the robotics, focusing on collision avoidance and trajectory planning for mobile robots (Thorpe and Matthies, 1984). In the automotive domain, APF was first applied to path planning and dynamic steering control for autonomous vehicles, successfully achieving collision-free navigation in simulations (Krogh and Thorpe, 1986). The basic APF model consists of an attractive potential for target points and a repulsive potential for obstacles (Khatib, 1986).

Subsequent studies improved this model by considering the influences of lane markings and road boundaries (Gerdes and Rossetter, 2001). By establishing repulsive potentials near road edges and lane markings while setting the potential at the lane center to zero, vehicles can be constrained to operate within the lane, leading to the development of lane-keeping systems. Additionally, other researchers integrated speed potentials and vehicle potentials into the model (Wolf and Burdick, 2008). When a vehicle's velocity is below the desired level, it experiences forward thrust, while exceeding the desired speed results in resistance. The presence of other vehicles on the road also generates repulsive potentials, guiding the vehicle's decision to follow or change lanes based on the magnitudes of these potentials.

Further advancements considered the future motion states of surrounding vehicles, categorizing them into those deviating from their lanes and those maintaining in the current lanes, each with tailored potential field functions (Sattel and Brandt, 2008). Subsequently, the driving safety field was proposed, which accounts for the interactions among drivers, vehicles, and environment, enabling a quantitative assessment of various risks in complex driving scenarios. In summary, the APF approach demonstrates significant potential for omnidirectional risk assessment, and if combined with consideration of vehicles' intention uncertainty, it may facilitate the development of an accurate and robust omnidirectional safety metric.

To represent the intention uncertainty of vehicles, trajectory prediction serves as a primary method. Accurate prediction of future motions allows autonomous vehicles to better understand their surrounding environment, thereby improving the quality and safety of decision-making. Existing trajectory prediction approaches can be broadly categorized into traditional model-based methods and learning-based methods. Traditional approaches often rely on kinematic models, physical constraints, or handcrafted rules (Berthelot et al., 2011; Althoff and Mergel, 2011; Gao et al., 2020b), which offer interpretability but typically struggle to generalize in complex, real-world driving scenarios.

In contrast, deep learning-based methods have gained increasing popularity due to their ability to capture complex patterns from large-scale datasets. These methods include both deterministic and probabilistic frameworks. Deterministic models such as Social-GAN (Gupta et al., 2018), which introduces adversarial training to generate socially acceptable trajectories, and VectorNet (Gao et al., 2020a), which models map and agent interactions using vector representations, have shown strong performance. Probabilistic models such as MultiPath (Chai et al., 2020) and MTP (Cui et al., 2019) focus on multimodal output generation, capturing multiple possible future paths conditioned on past observations. These approaches are particularly well-suited for highly interactive and uncertain scenarios.

More recently, several advanced deep learning models have achieved strong performance by incorporating interaction modeling and rich contextual understanding. QCNet (Zhou et al., 2023) introduces a spatio-temporal attention mechanism through a query-centric encoding scheme, enabling the model to better capture historical interactions and future uncertainties. ForecastMAE (Cheng et al., 2023) adopts a masked autoencoding architecture to enhance the model's robustness in encoding multi-lane traffic information and spatial layouts. MTR (Shi et al., 2024) leverages a transformer-based decoder with motion priors to achieve high-quality and diverse trajectory predictions. LaneRCNN (Zeng et al., 2021) explicitly models the spatial relationship between agents and lane structures, effectively learning driving intentions through scene-aware trajectory filtering. GANet (Wang et al., 2023) introduces a goal-aware prediction framework that jointly optimizes motion forecasting and destination estimation, demonstrating strong performance in both accuracy and diversity.

The availability of large-scale trajectory datasets such as Argoverse 2, Waymo Open Motion Dataset, and nuScenes has further facilitated the advancement and benchmarking of these predictive models.

In this work, our focus is not on proposing new trajectory prediction methods, but rather on utilizing the results of multimodal trajectory prediction to enable more accurate omnidirectional risk assessment. Consequently, we will employ open-source trajectory prediction methods and structure the trajectory prediction component as a separate module, allowing for future module replacements if necessary.

3. Model of interactive risk

In this section, we will provide a detailed introduction to the modeling process of IR. First, we utilize an open-source multimodal trajectory prediction method to generate multiple predicted trajectories along with their associated probabilities. Next, we construct cross-sections along a predicted trajectory using Gaussian distribution to model the risk distribution associated with that trajectory. Then, we integrate the risk distributions associated with each predicted trajectory, combining them with the vehicle's virtual mass to develop the IRDF model and propose a method for determining risk directions at various locations. Finally, we introduce the omnidirectional safety metric for quantifying the interactive risk between vehicles — IR. The framework of the modeling process is illustrated in Fig. 2.

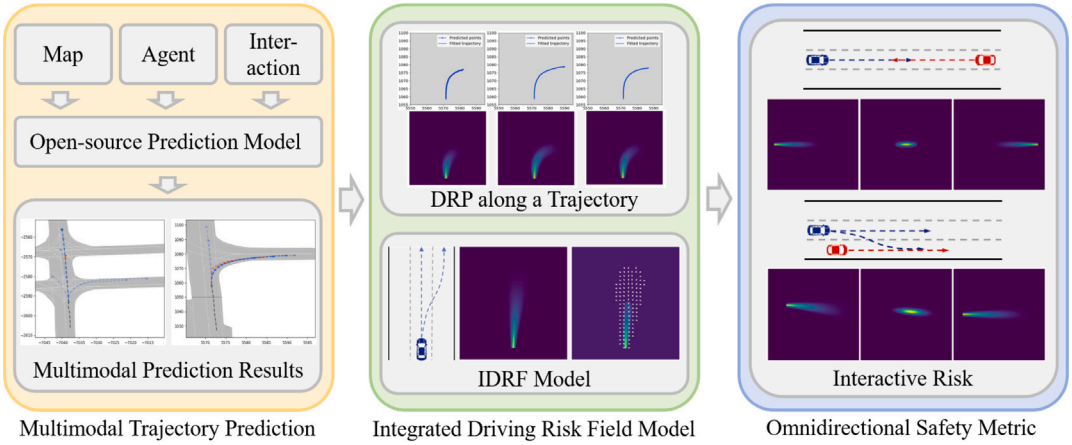


Fig. 2. The framework of Integrated Driving Risk Field (IDRF) and Interactive Risk (IR).

Table 1
Precision of trajectory prediction models on the Argoverse 2 dataset.

Models	minADE	minFDE
QCNet	0.62	1.19
ForecastMAE	0.72	1.43
SEPT	0.61	1.15
MTR	0.73	1.44
MTR++	0.71	1.37
GANet	0.72	1.34
LaneRCNN	0.87	1.36

3.1. Multimodal trajectory prediction

We mainly utilized QCNet (Zhou et al., 2023) as the multimodal trajectory prediction model, which is trained on the Argoverse 2 dataset. The reasons for selecting QCNet are as follows: (1) Although it is not currently the state-of-the-art (SOTA), QCNet is widely recognized as a representative model in trajectory prediction, with minimal performance differences compared to SOTA models (a comparasion of QCNet with several recent models is summarized in Table 1); (2) Given the inherent uncertainty in trajectory prediction, QCNet's output — comprising multiple predicted trajectories along with their associated probabilities — adequately meets our needs. To further evaluate the robustness of the proposed IR metric, we additionally conducted experiments using ForecastMAE as an alternative trajectory prediction model. The corresponding results are presented in Section 4. As mentioned earlier, enhancing the accuracy of the trajectory prediction module is not the primary focus of our research. If more performant models emerge in the future, we can easily implement a straightforward module replacement. Some examples of trajectory prediction using QCNet is illustrated in Fig. 3. The model is capable of predicting various future trajectory modalities, including acceleration or deceleration, lane keeping or lane changing, and straight or turning maneuvers. In Fig. 3, the dark gray dashed line represents 50 frames (5 s) of historical data, the blue dashed lines show 6 predicted trajectories for 60 frames (6 s), with deeper blue indicating higher probabilities, and the red dashed line represents the ground truth of the predicted trajectory.

3.2. Risk distribution along a predicted trajectory

The driver's risk field was proposed in Kolekar et al. (2020) and modeled as a torus with a Gaussian cross-section, where the shape of the torus was determined by the vehicle's current steering angle and velocity, following either a straight line or an arc. We expanded this model in Jiang et al. (2024) by replacing the kinematic-based straight line or arc with a predicted trajectory, incorporating future vehicle motion to provide a more comprehensive representation of risk. The risk distribution along a predicted trajectory is modeled within the Frenet coordinate system and subsequently transformed into the global Cartesian coordinate system. Taking the vehicle's current position as the origin, a Frenet coordinate system is established based on the predicted trajectory, along which the height of the Gaussian distribution decreases and its width increases. In practical calculations, the Frenet coordinates (s, d) corresponding to a given point (x, y) are first determined, enabling the risk distribution at that location to be computed based on the Gaussian distribution:

$$\pi^C(x, y) = \pi^F(s, d) = h(s) \cdot \exp\left(-\frac{d^2}{2w(s)^2}\right) \tag{1}$$

where π^C is the risk distribution at the point (x, y) in the Cartesian coordinate system, π^F is the risk distribution at the corresponding point (s, d) in the Frenet coordinate system. h and w represent the height and width of the Gaussian distribution respectively, and the variations are defined as follows:

$$h(s) = a \cdot (s - s_{\text{traj}})^2 \tag{2}$$

$$w(s) = (b + k \cdot \bar{\kappa}_{\text{traj}}) \cdot s + c \tag{3}$$

where s_{traj} and $\bar{\kappa}_{\text{traj}}$ represent the length and average curvature of the predicted trajectory, and a, b, c, k are constant coefficients. As s increases, the height h of the Gaussian distribution decreases quadratically, while the width w increases linearly. The larger the average curvature $\bar{\kappa}_{\text{traj}}$, the more pronounced the increase in w as s grows. This aligns with human intuition that, the uncertainty along the predicted trajectory becomes progressively stronger.

To clearly illustrate the risk distribution along a trajectory, we use a segment of a sinusoidal curve as the predicted trajectory. Fig. 4 demonstrates the establishment of the Frenet coordinate system in this scenario, as well as the variation of the Gaussian cross-section with respect to the prediction distance s. We also employ the multimodal trajectory prediction results from the dataset to generate the risk distributions corresponding to each predicted trajectory, which are presented in Fig. 5 alongside their probabilities.

3.3. Integrated driving risk field

By weighting the risk distributions of each predicted trajectory according to their respective probabilities and then summing them up,

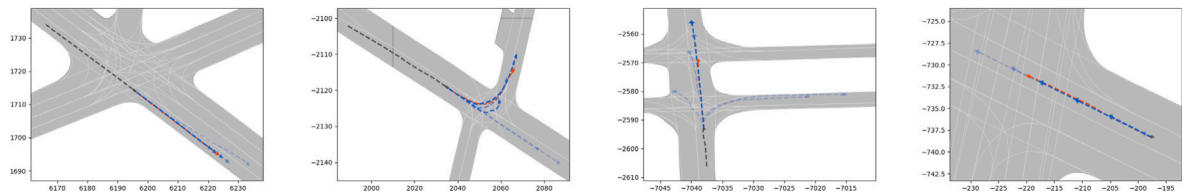


Fig. 3. Some examples of trajectory prediction using QCNet (Zhou et al., 2023). The results may contain multiple modalities, such as acceleration or deceleration, lane keeping or lane changing, and straight or turning maneuvers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

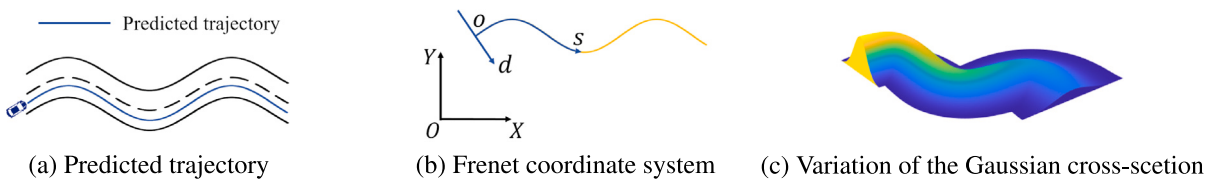


Fig. 4. The establishment of the Frenet coordinate system and the variation of the Gaussian cross-section when the predicted trajectory is a sinusoidal curve.

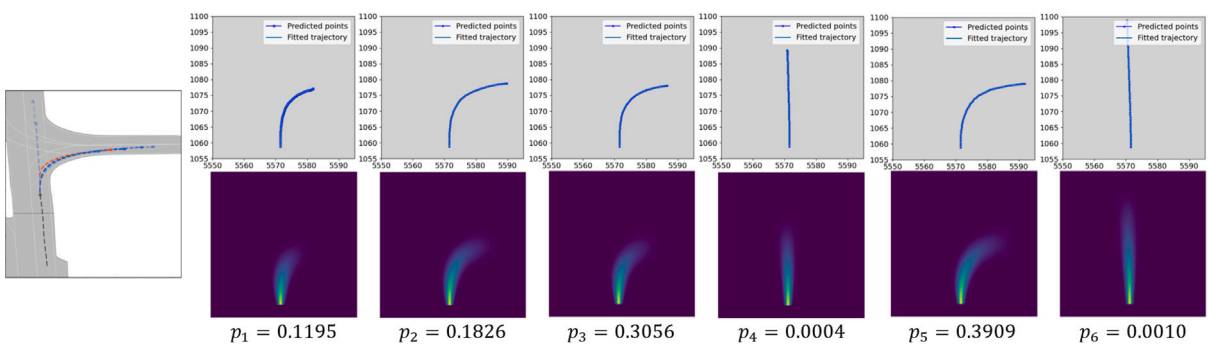


Fig. 5. The multimodal prediction results for a specific scenario in the dataset, along with the risk distributions and the probabilities associated with each predicted trajectory.

the integrated risk distribution can be obtained.

$$\pi_I^C(x,y) = \sum_{i=1}^{n} p_i \cdot \pi^C(x,y) = \sum_{i=1}^{n} p_i \cdot \pi^F(s_i,d_i) = \sum_{i=1}^{n} p_i \cdot h(s_i) \cdot \exp\left(-\frac{d_i^2}{2w(s_i)^2}\right) \tag{4}$$

where π_I^C is the integrated risk distribution, p_i is the probability of the ith predicted trajectory, and (s_i, d_i) are the Frenet coordinates corresponding to the point (x, y) on the ith predicted trajectory.

Based on traffic risk quantification theory, accident risk is typically modeled as the product of the probability of an accident occurring and the consequences of that accident (Näätänen and Summala, 1976). π_I^C can represent the probability component of traffic risk; therefore, it is essential to also model the consequence of an accident, which is related to the current state of the vehicle (including factors such as mass, type, size, and velocity). In Wang et al. (2015a), through the analysis of natural driving data, the influence of vehicle state on traffic accident consequence is summarized as *Virtual Mass*, which is defined as:

$$M_V = m \cdot T \cdot (\alpha \cdot v^{\beta} + \gamma) \tag{5}$$

where M_V represents the virtual mass, and m, T, and v are the vehicle's mass, type, and current velocity respectively. The parameters α, β, and γ are constant coefficients determined through regression on empirical accident data. Their specific values are listed in Table 2. Notably, α is a positive constant and $\beta > 1$, which implies that the consequence of an accident increases rapidly with vehicle speed. This behavior is consistent with empirical observations and aligns with common safety intuition in traffic scenarios.

Multiplying π_I^C, which represents the accident probability component, by M_V, which represents the accident consequence component, yields the final IDRF model:

$$\mathrm{IDRF}(x,y) = \pi_I^C(x,y) \cdot M_V = m \cdot T \cdot (\alpha \cdot v^{\beta} + \gamma) \cdot \sum_{i=1}^{n} p_i \cdot h(s_i) \cdot \exp\left(-\frac{d_i^2}{2w(s_i)^2}\right) \tag{6}$$

The above process quantifies the magnitude of risk at various locations around the vehicle; however, the direction of that risk is also important, as it influences the intensity of potential collisions when two vehicles may intersect at a specific point. For example, in the scenarios illustrated in Fig. 6(a) and (b), the magnitudes of the IDRF for both vehicles at point Q may be identical, yet it is evident that the risk of possible head-on collision is greater than that of possible rear-end collision. This highlights the impact of risk direction surrounding the vehicle. The direction of risk at a specific location is defined as the gradient direction of the IDRF at that point, which can be expressed as:

$$\mathbf{r}(x,y) = \frac{\nabla \mathrm{IDRF}(x,y)}{\|\nabla \mathrm{IDRF}(x,y)\|} \tag{7}$$

In summary, the IDRF is determined by the multimodal trajectory prediction results of the vehicles and their states. The parameters of the IDRF include $\alpha, \beta, \gamma, a, b, c, k$, with their respective values shown in Table 2. These parameter values are chosen by referring to prior studies on potential field-based risk modeling, particularly Kolekar et al. (2020) and Wang et al. (2016), which provide empirically validated parameter ranges for similar driving risk field formulations. We visualize the IDRF results (including risk magnitude and direction) on different example scenarios, as illustrated in Fig. 7. Among these, (a) and (b) represent

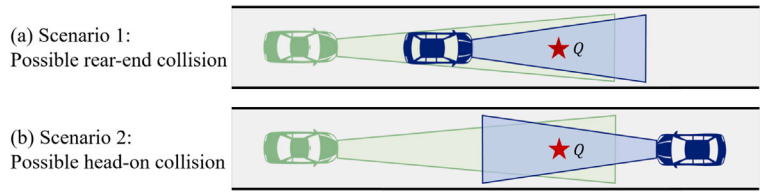


Fig. 6. Illustration of impact of risk direction. In these two scenarios, although the magnitudes of IDRF generated by the two vehicles at point Q are the same, it is evident that the risk of scenario 2 (a possible head-on collision) is greater than that of scenario 1 (a possible rear-end collision).

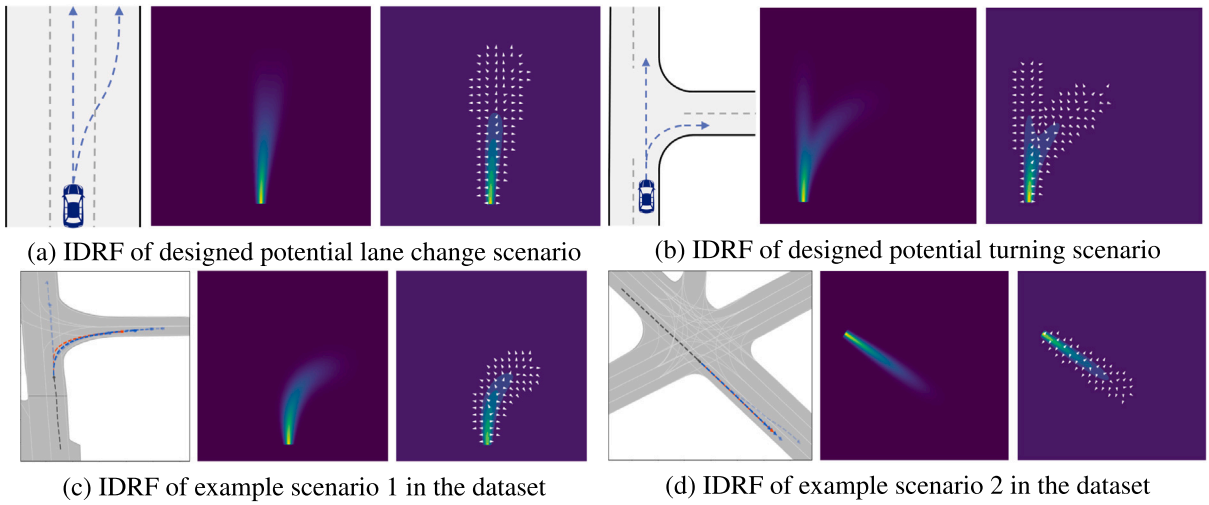


Fig. 7. The IDRF results (including risk magnitude and direction) on different example scenarios. (a) and (b) Represent the designed potential lane change and turning scenarios, while (c) and (d) depict scenarios from the dataset.

Table 2
Parameters in the IDRF.

α	β	γ	a	b	c	k
1.566×10^{-14}	6.687	0.3345	0.001	0.04	0.5	1

the designed potential lane change and turning scenarios, while (c) and (d) depict scenarios from the dataset. It is clear that the IDRF can effectively represent the uncertainty of vehicle intentions and the surrounding risk distribution, thereby providing the foundation for omnidirectional risk assessment in interactive scenarios.

3.4. Interactive risk

In real-world traffic environments, risk emerges from complex interactions among multiple entities. However, since most traffic accidents can ultimately be attributed to conflicts between pairs of agents, our formulation primarily focuses on quantifying the interaction risk between two entities. The risk associated with multi-agent interactions can thus be decomposed into a series of pairwise interaction risks. Based on the IDRF and the direction of risk, the IR between any two vehicles i and j can be quantified, which is defined as:

$$\mathrm{IR}(x, y) = \mathrm{IDRF}_i(x, y) \cdot \mathrm{IDRF}_j(x, y) \cdot n^{-\mathbf{r}_i(x,y)\cdot\mathbf{r}_j(x,y)} \tag{8}$$

Here, we use $n^{-\mathbf{r}_i(x,y)\cdot\mathbf{r}_j(x,y)}$ to model the influence of the risk direction and n is a parameter greater than 1. If the IDRF values generated by vehicles i and j at location (x, y) are identical, the risk is minimized when they move in the same direction and maximized when they move in opposite directions. This aligns with the intuitive perception of risk by humans. To better understand (8), the IDRF can be decomposed according to (6), as follows:

$$\mathrm{IR}(x, y) = \pi^C_{I,i}(x, y) \cdot \pi^C_{I,j}(x, y) \cdot M_{V,i} \cdot M_{V,j} \cdot n^{-\mathbf{r}_i(x,y)\cdot\mathbf{r}_j(x,y)}$$

where $\pi^C_{I,i}(x, y) \cdot \pi^C_{I,j}(x, y)$ can be interpreted as the probability of vehicles i and j simultaneously being at location (x, y) and thus colliding. Meanwhile, $M_{V,i} \cdot M_{V,j} \cdot n^{-\mathbf{r}_i(x,y)\cdot\mathbf{r}_j(x,y)}$ can be seen as the consequence of a collision occurring between i and j at (x, y), representing its severity. Therefore, the model for IR accords with the fundamental model for quantifying traffic risk. Ultimately, the safety metric used to assess the risk level between the two vehicles, is defined as the maximum value of the IR across all locations, that is:

$$\mathrm{IR}_{\max} = \max_{(x,y)} \mathrm{IR}(x, y) \tag{9}$$

We use a potential head-on collision scenario (Fig. 8(a)) and a cut-in scenario (Fig. 8(b)) to demonstrate the IR. The vehicles' respective IDRFs and the IR during the interaction are also shown in Fig. 8.

4. Model verification

In this section, we conduct experimental validation of the proposed safety metric IR. Firstly, we designed four typical collision scenarios: head-on collision, adjacent vehicle cut-in collision, angle collision, and collision during turning at an intersection. In these scenarios, we compare the performance of IR with other baseline safety metrics. Subsequently, we further validate the effectiveness of IR through data from dangerous scenarios in driving simulator experiments. Finally, we evaluate the performance of IR in more complex multi-vehicle interaction scenarios by selecting some high-risk cases from the Argoverse 2 dataset, which consists of real-world driving data collected by autonomous vehicles.

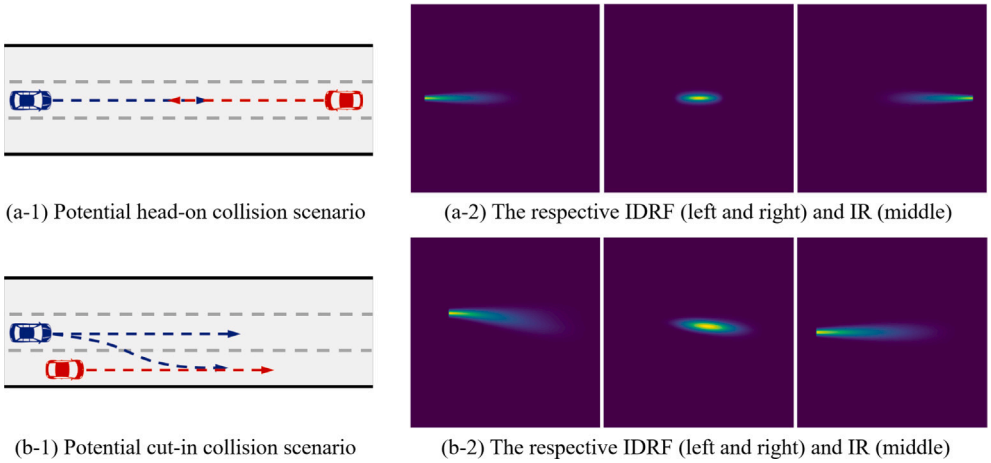


Fig. 8. The designed typical scenarios, the IDRF for each vehicle and the IR during the interaction.

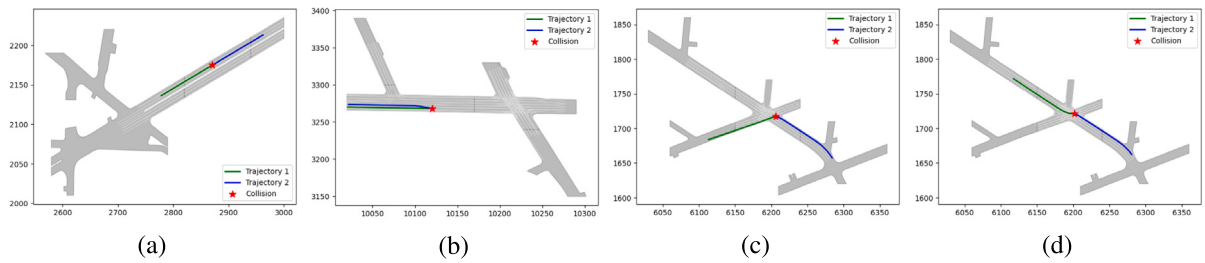


Fig. 9. The designed typical collision scenarios. (a) Is a head-on collision between two vehicles traveling in opposite directions; (b) is a collision caused by one vehicle suddenly cutting into the lane of another vehicle when both are traveling nearly parallel; (c) is an angle collision between two vehicles traveling perpendicularly at an intersection; and (d) is a collision between a left-turning vehicle and a vehicle traveling straight through an intersection.

4.1. Case study of typical collision scenarios

Here, we design four typical collision scenarios: (1) a head-on collision between two vehicles traveling in opposite directions, (2) a collision caused by one vehicle suddenly cutting into the lane of another vehicle when both are traveling nearly parallel, (3) an angle collision between two vehicles traveling perpendicularly at an intersection, and (4) a collision between a left-turning vehicle and a vehicle traveling straight through an intersection. Each of the two vehicles follows a trajectory lasting 9.5 s, with trajectory points recorded every 0.1 s, and the collision occurs at the final point. The trajectories and the collision points for each scenario are shown in Fig. 9.

To comprehensively evaluate the performance of the IR, we selected several baseline safety metrics for comparison, including two-dimensional TTC (2D-TTC, Guo et al., 2023), anticipated collision time (ACT, Venthuruthiyil and Chunchu, 2022), time advantage (TAdv, Laureshyn et al., 2010), and emergency index (EI, a recently proposed two-dimensional safety metric (Cheng et al.)). Using the start of each vehicle's trajectory as the reference time (0 s), the values of each safety metric were recorded from 5 s to 9.5 s (when the collision occurs) for each scenario. The results are shown in Fig. 10.

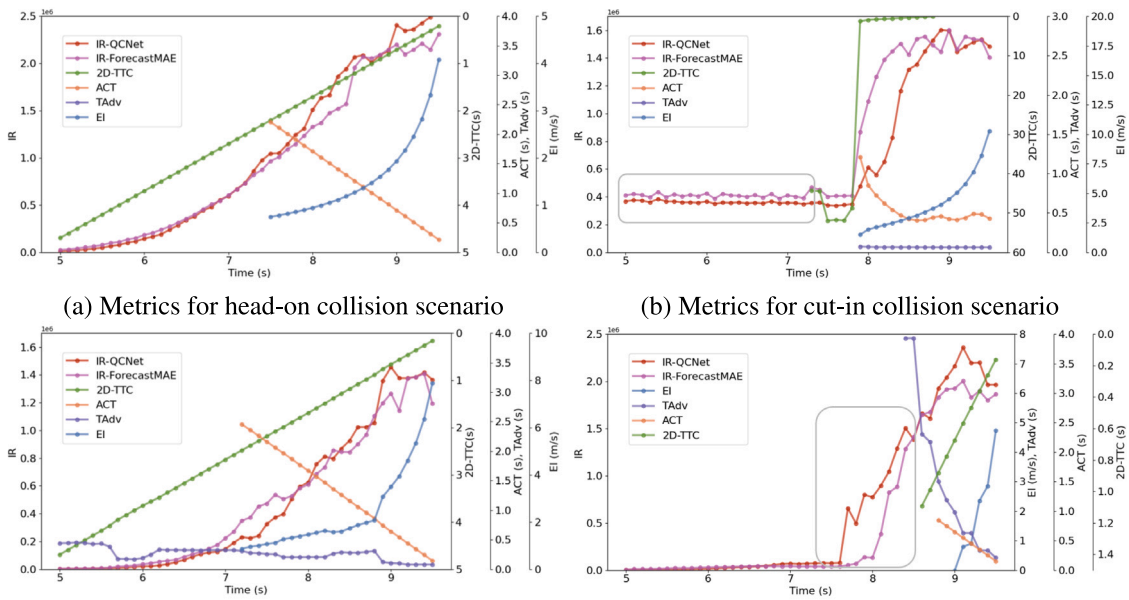
In order to further validate the robustness of IR under different trajectory prediction frameworks, we introduce an additional baseline using the ForecastMAE model, an open-source and recently proposed motion prediction method capable of generating multimodal trajectory distributions. As shown in Fig. 10, both IR-QCNet and IR-ForecastMAE exhibit consistent trends in all four designed collision scenarios, confirming that IR is effective regardless of the specific prediction module used. Notably, despite some variations in absolute values — due to differences in predicted trajectory distributions — both versions of IR consistently detect potential risks earlier than conventional safety metrics (e.g., 2D-TTC, EI, ACT, and TAdv), particularly in complex interactions such as cut-in and intersection collisions. This consistency across models reinforces the reliability of IR as a generalizable risk metric, rather than one dependent on a specific trajectory predictor.

Scenario 1: head-on collision (Fig. 10(a)). In this scenario, both IR and 2D-TTC detect the collision risk at $t = 5$ s, while EI and ACT only start to respond at around $t = 7.5$ s. Since a head-on collision is a one-dimensional scenario, 2D-TTC (simplified to one-dimensional TTC here) performs effectively. However, unlike the linear trend of 2D-TTC, IR exhibits an exponential-like increase, with a higher rate of change closer to the collision moment. This pattern aligns more closely with human intuition of risk perception. TAdv is ineffective for scenario of head-on collision.

Scenario 2: cut-in collision (Fig. 10(b)). At $t = 7.5 - 8$ s, one vehicle suddenly begins to cut into the adjacent lane, triggering all safety metrics. A distinct advantage of IR is its ability to maintain a certain risk level even before the lane change maneuver, when the two vehicles are traveling nearly parallel. This aligns with human intuition, as parallel driving also poses a potential risk. However, other safety metrics fail to capture this effect.

Scenario 3: angle collision (Fig. 10(c)). This scenario is essentially similar to a head-on collision, as both involve two vehicles traveling along their respective linear trajectories and colliding. Therefore, the trends of the safety metrics in this case are also basicly similar to those observed in the head-on collision. Both IR and 2D-TTC can detect collision risks earlier than EI and ACT, while TAdv has little effectiveness in this regard.

Scenario 4: intersection collision (Fig. 10(d)). This is the most challenging scenario among the designed cases. Baseline safety metrics generally assume constant linear motion at each moment, making it difficult to detect collision risks in scenarios involving a turning vehicle and a straight-moving vehicle. In contrast, IR leverages the results of multimodal trajectory prediction for risk assessment, allowing it to



(a) Metrics for head-on collision scenario (b) Metrics for cut-in collision scenario

(c) Metrics for angle collision scenario (d) Metrics for collision at an intersection scenario

Fig. 10. The comparative results of different safety metrics across the four designed scenarios. (a) For head-on collision, (b) for cut-in collision, (c) for angle collision and (d) for collision between a left-turning vehicle and a vehicle traveling straight through an intersection.

anticipate potential turning behaviors and achieve better risk evaluations. As shown in Fig. 10(d), IR detects collision risk approximately 1 s earlier than other safety metrics, which is crucial in emergency situations.

Additionally, it can be observed that the peak values of IR vary across different scenarios, reflecting the influence of risk direction. In head-on and intersection collisions, where the vehicles approach nearly head-on, the potential consequences are more severe, resulting in higher IR values compared to the side-collision scenarios like cut-in and angle collisions. Unlike other kinematics-based safety metrics, IR may show slight decreases at certain moments, occasionally deviating from the overall trend. This is due to the multimodal trajectory prediction used in IR; sudden changes in predicted trajectories — such as an abrupt appearance and then disappearance of lane-change or turning maneuvers — can cause momentary drops in IR. However, this does not detract from IR's overall alignment with human intuition of collision risk. With further advancements in the trajectory prediction module, it is anticipated that IR's variation curve will become smoother, with fewer or no anomalies.

In summary, compared to baseline safety metrics, IR detects collision risks earlier and responds to risks in a way that more closely aligns with human intuition.

4.2. Validation through dataset

After the case study of typical collision scenarios, we further validate IR's capability to represent risk using data of the hazardous scenario. Here, we utilize the D2E dataset (Ke et al., 2024), an autonomous decision-making dataset that was collected via a driving simulator. We use the scenario — car following with preceding car braking on urban roads (as shown in Fig. 11) — from the D2E dataset, as an example to examine the variation patterns of IR under instances of successful and unsuccessful collision avoidance.

Since the D2E dataset does not model map information, we first model the lane markings, road boundaries, and other relevant features in the scenario to enable more accurate trajectory prediction. Then, for different instances, we calculate the IR between the ego vehicle and the preceding vehicle, and show its variation over time. Fig. 12 illustrates the changes in IR under instances of successful and unsuccessful collision avoidance. To better highlight the characteristics of IR, we also mark the moments of sudden braking by the preceding vehicle (represented by red dashed lines) in each subfigure of Fig. 12.

It can be observed that, in instances of collision occurrence, the peak value of IR closely corresponds to the moment of braking by the preceding vehicle, indicating that the ego vehicle did not anticipate the risk in advance. In contrast, in instances of successful collision avoidance, the peak value of IR occurs prior to the braking moment of the preceding vehicle, suggesting that the ego vehicle detected the potential danger early and kept the risk within a controllable range. The differing temporal variation of IR in instances of successful and unsuccessful collision avoidance further demonstrates its ability to reflect the risk.

4.3. Evaluation in multi-vehicle interaction scenarios

To further evaluate the robustness and applicability of the proposed IR metric in complex real-world environments, we selected three high-risk scenarios involving multiple interacting vehicles from the Argoverse 2 dataset. Unlike the controlled two-vehicle collision cases in Section 4.1, these real-world cases are characterized by dense traffic, multiple interaction intents, and diverse driving maneuvers.

As illustrated in Fig. 13 (a-1), (b-1) and (c-1), each subfigure presents a multi-vehicle interaction scenario from three perspectives: a global view at $T = 0$ s (left), and two local close-up views at $T = 1$ s and $T = 4$ s (middle and right). In each scenario, a specific target vehicle is selected for IR evaluation — vehicle 0 in scenario 1, vehicle 1 in scenario 2, and vehicle 3 in scenario 3. For each target vehicle, we compute and track its IR values with respect to surrounding vehicles throughout the interaction process. These IR interactions are denoted as IR-A-B, where A refers to the index of the selected target vehicle and B refers to the index of another interacting vehicle. The corresponding IR curves over time are shown in Fig. 13 (a-2), (b-2) and (c-2), capturing the dynamic evolution of risk as the scenarios unfold.

Scenario 1: lane change and overtaking at an intersection (Fig. 13(a)). This scenario takes place at a complex urban intersection involving multiple traffic participants. After passing through the intersection, vehicle 0 (the target vehicle) initiates a lane change maneuver to overtake vehicle 4. As shown in Fig. 13 (a-2), around $T = 1$ s, vehicle

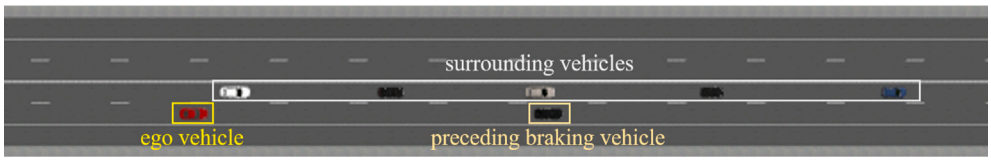


Fig. 11. Illustration of the designed scenario (car following with preceding car braking on urban roads) in D2E dataset.

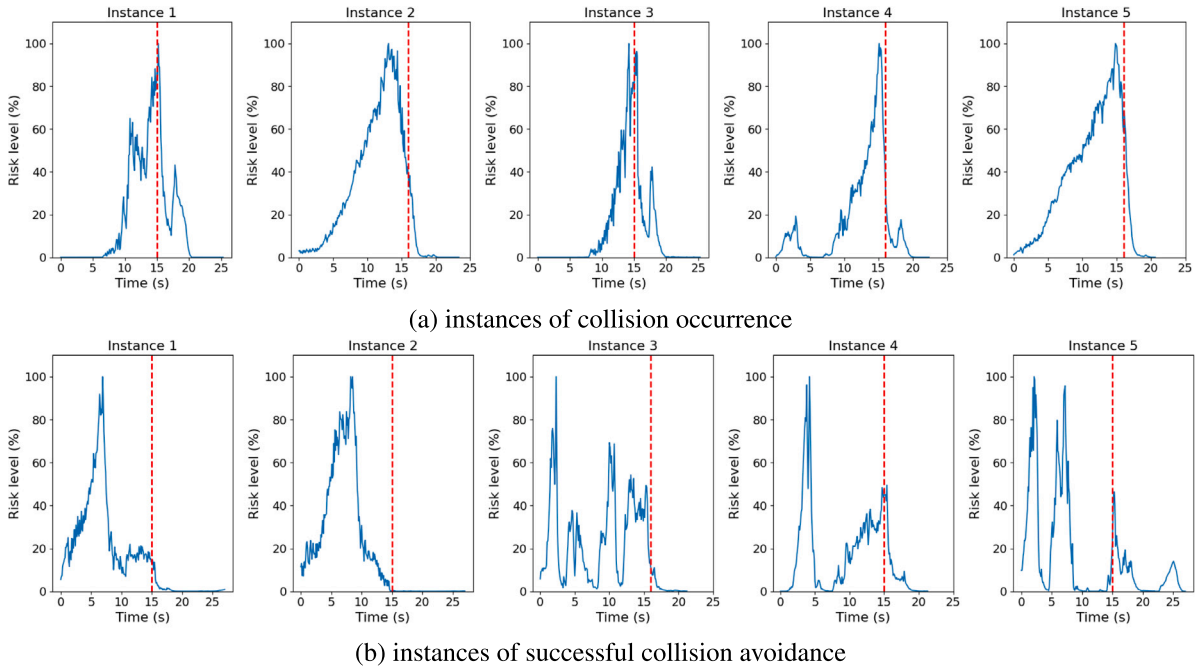


Fig. 12. The variation of IR over time in instances of successful and unsuccessful collision avoidance.

0 starts moving toward the adjacent lane, leading to a peak value in IR-0-4, indicating a heightened collision risk due to potential trajectory overlap. By $T = 4$ s, vehicle 0 has completed the lane change and drives almost side-by-side with vehicle 4. At this point, both vehicles maintain stable trajectories without further lateral movements, and the IR value decreases accordingly, though it remains at a moderate level reflecting residual proximity-based risk. The IR values between vehicle 0 and other surrounding vehicles remain close to zero throughout, suggesting minimal interaction or conflict.

Scenario 2: parallel driving with close longitudinal and lateral distance (Fig. 13(b)). This scenario occurs just after an intersection, where vehicle 1 (target vehicle) moves alongside vehicles 0, 2, and 4 in the same general direction. Specifically, vehicle 0 follows closely behind in the same lane, vehicle 2 travels side-by-side in the right adjacent lane, and vehicle 4 is ahead of vehicle 0 in another lane. As illustrated in Fig. 13(b-2), IR-1-0 reaches the highest value due to the close-range following situation. IR-1-2 also remains high, reflecting the sustained lateral proximity and potential for side collision despite no current lane-changing behavior. IR-1-4 is comparatively low, as the two vehicles are separated both laterally and longitudinally. Throughout the sequence, the relative positions between vehicles remain stable, which limits the responsiveness of traditional safety metrics that rely on dynamic changes. However, IR effectively captures the latent risk arising from close spacing and high speeds, highlighting its advantage in evaluating subtle but critical multi-agent interactions.

Scenario 3: lane-changing risk during high-speed overtaking (Fig. 13(c)). This scenario unfolds on a straight road segment, where vehicle 3 (target vehicle) and surrounding vehicles 1, 2, 4, and 5 are all moving in the same direction. As shown in Fig. 13(c-2), vehicle 3 accelerates during the interaction, transitioning from behind vehicles 1 and 2 to overtaking them. Laterally, vehicles 1 and 2 are located in the adjacent left and right lanes, respectively, while vehicle 4 is two lanes to the left, and vehicle 5 follows vehicle 1. As vehicle 3 picks up speed, the trajectory prediction model indicates a possible left lane change, leading to a higher IR value between vehicles 3 and 1. In contrast, the IR between vehicles 3 and 2 remains lower, reflecting weaker lateral interaction. Due to greater lateral and longitudinal distances, interactions with vehicles 4 and 5 show only minimal risk.

Notably, IR values in these scenarios fluctuate more significantly over time than in the manually designed cases in Section 4.1. This is primarily due to the variability in vehicle speeds within real-world data, which causes the virtual mass to change over time. Additionally, the trajectory prediction module introduces increased temporal noise in forecasting vehicle behavior, further amplifying frame-to-frame IR variations.

These cases demonstrate that IR effectively captures subtle and evolving risks in complex multi-vehicle interactions, offering a level of responsiveness and foresight that traditional safety metrics often fail to achieve in real-world scenarios.

5. Model application

Based on the model of IR, there are four cotegories of applications as follows:

(1) Traffic risk monitoring. This can be achieved by deploying roadside sensor devices along road sections or at intersections to continuously record traffic data. Over a specified period, the collected data can be analyzed to compute the IR at various locations within the scene, which provides an indication of the relative risk at each point. By evaluating the spatial distribution of IR values, this approach

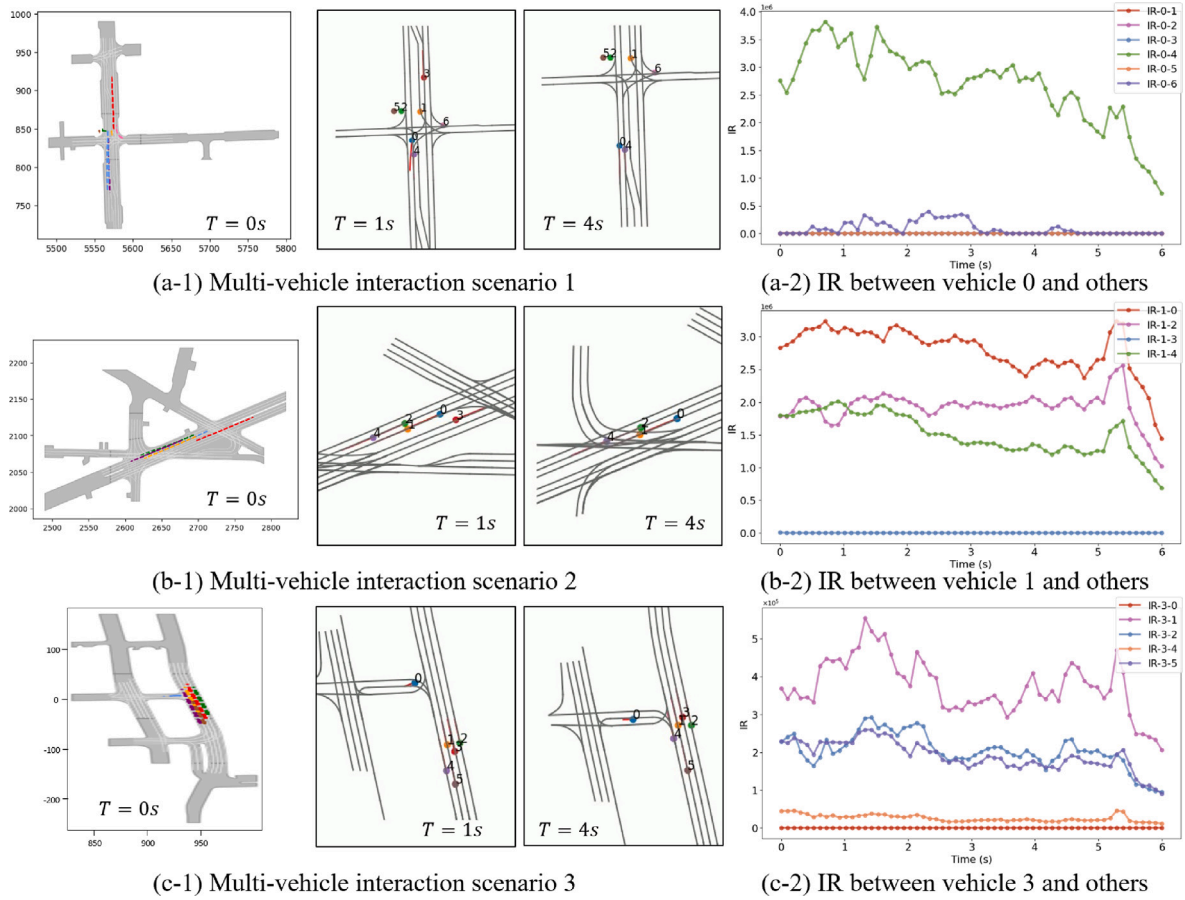


Fig. 13. Evaluation of IR in three complex multi-vehicle interaction scenarios from the Argoverse 2 dataset. Each row shows a scenario at $T = 0$ s, $T = 1$ s, and $T = 4$ s (left to right), and the rightmost plot shows the IR value over time between the target vehicle and surrounding agents.

can inform decision-making processes related to road planning and the strategic placement of traffic signs and signals. Specifically, locations with higher average IR values are indicative of areas with a greater likelihood of traffic accidents, suggesting that targeted interventions, such as the installation of traffic signals, warning signs, or other safety measures, could be effective in mitigating the risk and reducing the probability of accidents.

(2) Ego-vehicle risk warning. The ego-vehicle risk warning system can leverage a combination of sensors installed on the ego vehicle, such as cameras and LiDAR, or utilize Vehicle-to-Vehicle (V2V) communication to continuously gather real-time kinematic information about surrounding vehicles. Using the data collected, along with trajectory prediction algorithms, the system calculates the IR between the ego vehicle and other nearby vehicles. A risk threshold is then established, and if the IR exceeds this threshold, the system triggers an alert to warn the driver or the CAV of potential risk. This threshold can be dynamically adjusted based on the preferences of the driver or passengers, accommodating different driving styles, ranging from aggressive to neutral or conservative. By tailoring the warning system to the user's driving behavior, it ensures that the risk alerts are appropriately calibrated to enhance safety while minimizing unnecessary disruptions.

(3) Driving decision-making performance evaluation. The evaluation of driving decision-making is a critical area in the development of autonomous driving. Current mainstream methods typically involve a dimensional assessment approach, where decisions are evaluated across multiple dimensions, such as safety, time efficiency, comfort, energy efficiency, and other relevant factors. Each of these dimensions is assigned a specific weight, and a composite evaluation score is derived based on the weighted contributions of each dimension, which allows for a comprehensive evaluation from various perspectives. As an omnidirectional safety metric, IR can be integrated into any decision-making evaluation framework, serving as a basis for assessing the safety of decision outcomes and providing a more reliable measure of safety performance.

(4) Motion and trajectory planning. For the motion and trajectory planning of CAVs, IR can be utilized in various ways, depending on the specific framework of the planning module. One potential application is to use IR as a safety-checking criterion for the planned motion. In this case, if the IR of the planned motion exceeds a predetermined threshold, the planning process would be reconsidered or recalibrated to ensure the safety of the planned outcome. Another possible application is to use IR as a safety indicator for evaluating feasible planning trajectories. For instance, in a sampling-based trajectory planning framework, IR can be employed as one of the key safety indicators for the sampled trajectories. This would be combined with other relevant indicators, such as efficiency and smoothness, to select the optimal planned trajectory. Once the most suitable plan is chosen, it can be transmitted to the vehicle control module for execution, ensuring that both safety and other performance criteria are met.

6. Conclusions and future work

This paper proposed IR, an omnidirectional safety metric of CAVs based on multimodal trajectory prediction and driving risk field. Addressing the challenges of insufficient research on omnidirectional safety metrics and the lack of consideration for uncertainties arising

from vehicle intentions, IR combines the omnidirectional nature of the field with the capability of multimodal trajectory prediction to capture the uncertainty of vehicle intentions. It quantifies the probability and severity of potential accidents, thus providing a comprehensive measure of the risk level. According to the results from the case study of typical collision scenarios and experiments conducted using the dataset, IR can accurately reflect the risk level faced by CAVs, and detect collision risks earlier while respond in a manner that more closely aligns with human intuition when compared to baseline safety metrics. We have also proposed four key applications based on IR, including traffic risk monitoring, ego-vehicle risk warning, driving decision-making performance evaluation, and motion and trajectory planning.

Future work will explore integrating IR with other safety metrics, especially those that explicitly model temporal alignment, to further improve robustness and ensure safe decision-making in diverse traffic scenarios. We also plan to develop the applications of IR by fully integrating it into the risk warning and motion planning modules of CAVs. Additionally, efforts will be made to enhance the computational efficiency of the IR model in practical applications, thereby improving its overall practicability.

CRediT authorship contribution statement

Junkai Jiang: Visualization, Methodology, Writing – original draft, Validation, Investigation. **Zhiyuan Liu:** Validation, Data curation, Visualization, Methodology. **Hao Cheng:** Validation, Investigation, Methodology. **Zeyu Han:** Methodology, Visualization, Conceptualization. **Zehong Ke:** Validation, Investigation. **Yuning Wang:** Writing – review & editing, Supervision. **Qing Xu:** Project administration, Supervision, Conceptualization. **Jianqiang Wang:** Supervision, Conceptualization, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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